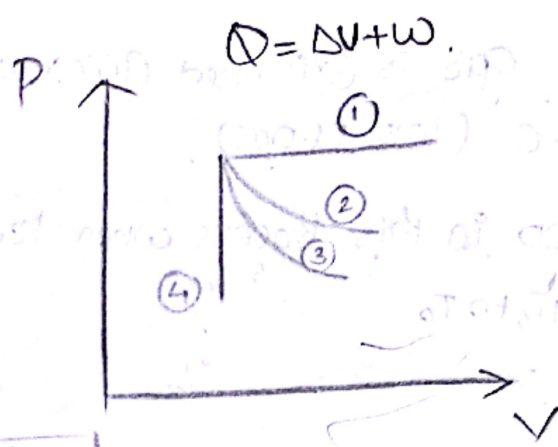




$$R_v^x = C$$

$$R_v^v = C$$

$$T_v^{1-\gamma} = C$$

$$P^{1-\gamma} T^\gamma = C$$

$$\gamma = 0 \text{ ISOBARIC}$$

$$\gamma = 1 \text{ ISOTHERMAL}$$

$$\gamma = \gamma \text{ ADIABATIC}$$

$$\gamma = \infty \text{ ISOCORIC}$$

$$R = C_p - C_v$$

$$\frac{C_p}{C_v} = \gamma$$

$$C = C_v + \frac{R}{1-\gamma}$$

PROCESS	γ	Q	ΔU	W	C
ISOBARIC	0	$nC_p \Delta T$	$nC_v \Delta T$	$nR \Delta T$	C_p
ISOTHERMAL	1	$nRT \ln(\frac{V_2}{V_1})$	0	$nRT \ln(\frac{V_2}{V_1})$	∞
ADIABATIC	γ	0	$nC_v \Delta T$	$\frac{nR \Delta T}{1-\gamma}$	0
ISOCORIC	∞	$nC_v \Delta T$	$nC_v \Delta T$	0	C_v

$$\frac{n_1 + n_2}{\gamma_{mix} - 1} = \frac{n_1}{\gamma_1 - 1} + \frac{n_2}{\gamma_2 - 1}$$

$$(C_v)_{mix} = \frac{n_1 C_{v1} + n_2 C_{v2}}{n_1 + n_2}$$

$$\ln p + \gamma \ln v = -\frac{\gamma}{\gamma-1} \ln \frac{p}{p_1}$$

$$\ln p + \ln \left(\frac{v}{v_1}\right) = C$$

$$\ln(p v^\gamma) = C$$

$$p v^\gamma = C$$

$$W = \frac{n R \Delta T}{\gamma - 1}$$

$$W = - \left[\frac{n R \Delta T}{\gamma - 1} \right]$$

$$W = - \left[n C_v \Delta T \right]$$

$$* * W = - \Delta U$$

$$* * \boxed{p v^\gamma = \text{Constant}}$$

$$W = \frac{n R \Delta T}{\gamma - 1}$$

$$\Delta U = n C_v \Delta T$$

$$Q = n C_p \Delta T$$

$$C = C_v + \frac{R}{\gamma - 1}$$

MOLAR SPECIFIC HEAT

$$Q = \Delta U + W$$

$$n C_p \Delta T = n C_v \Delta T + \frac{n R \Delta T}{\gamma - 1}$$

$$\boxed{C = C_v + \frac{R}{\gamma - 1}}$$

$\gamma = \frac{C_p}{C_v}$	$\frac{5}{3}$	$\frac{7}{5}$	$\frac{4}{3}$
C_p	$\frac{5R}{2}$	$\frac{7R}{2}$	$4R$
C_v	$\frac{3R}{2}$	$\frac{5R}{2}$	$3R$
GAS	Mono atomic ($f=3$)	Di ($f=5$)	Poly ($f=6$)

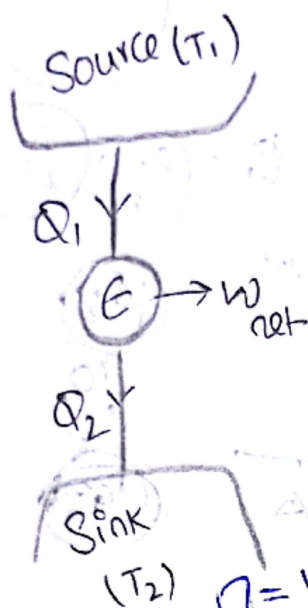
$$V(\text{internal Energy}) = \frac{n}{2} f R T$$

$$V = n C_v \Delta T$$

$$V = n C_v T$$

$$* * \boxed{C_v = \frac{f R}{2}}$$

HEAT ENGINE



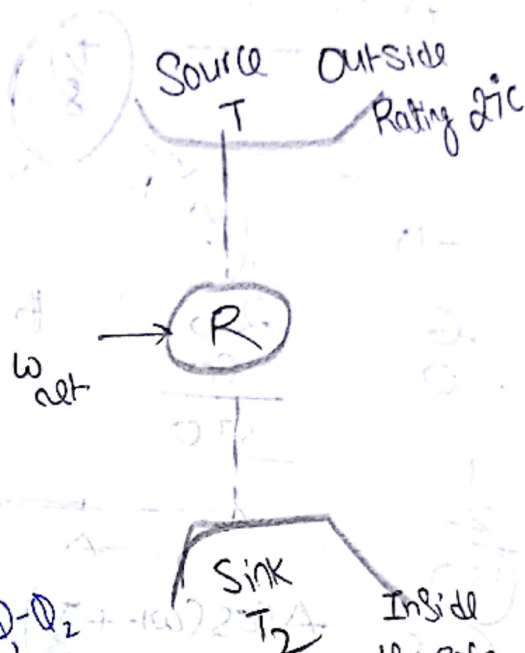
$$\eta = \frac{W_{net}}{Q_1} = \frac{Q_1 - Q_2}{Q_1}$$

Second Law of TD :

➔ It is impossible for any device

to transfer heat from low temp to high temp without any external work done on it.

REFRIGERATOR



$$COP = \frac{T_2}{T_1 - T_2}$$

* Coefficient of Performance (COP)

$$COP = \frac{\text{Desired Eff}}{\text{work i/p}}$$

$$COP = \frac{Q_2}{W_{net}}$$

$$COP = \frac{Q_2}{Q_1 - Q_2}$$

$$0.435 - 1 = 0.435$$

$$0.435 - 1 = 0.435$$

$$0.435 - 0.25$$

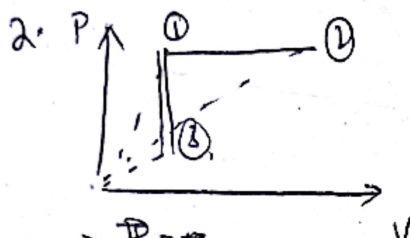
$$0.435 - 0.25$$



$$P_1 V_1 = nRT$$

$$V = \frac{T}{P} = \frac{1}{\text{slope}} \quad \left(V \propto \frac{1}{\text{slope}} \right)$$

Ans: $V_1 < V_2$



$$P_2 > P_1$$

$$P_2 > P_3$$

$$\Delta U = nC_V \Delta T$$

Thermodynamics:

• Study of heat...

Heat Transfer Work Transfer

$$Q = \Delta U + W$$

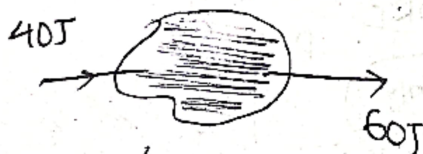
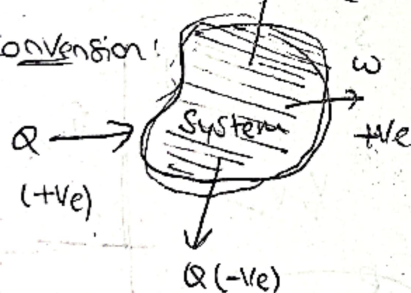
where

ΔU = Change in internal Energy

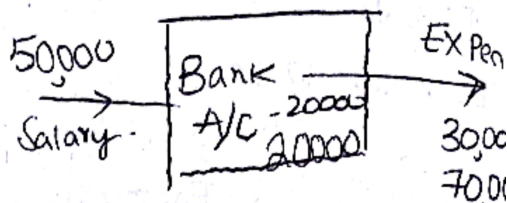
W = work Done

Q = heat

Sign convention:



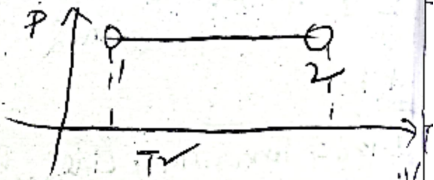
Adiabatic Process $Q = 0$



Q. Oxygen Gas at 10°C

$\Delta T \rightarrow$ Always in Kelvin...

1. Constant P Process:



$$W = \int P dV$$

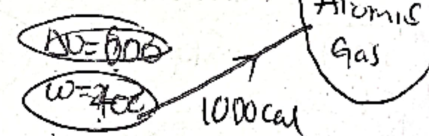
$$W = P \Delta V$$

$$W = nR \Delta T$$

$$Q = nC_P \Delta T$$

$$\Delta U = nC_V \Delta T$$

Q. constant Pressure



$$\Delta U = \frac{5}{2} \frac{1000}{3000} \left(\frac{5}{2} \right) C_P$$

constant pressure

4. Adiabatic Process:



Diatomic
Chm

$$W = 800 \text{ cal}$$

$$Q = 2800$$

$$\Delta U = 2000$$

$$Q = \Delta U + W$$

$$Q = nC_p \Delta T$$

$$\Delta U = nC_v \Delta T$$

$$2800$$



Diatomic
gas

$$\frac{Q = nC_p \Delta T}{\Delta U = nC_v \Delta T} = \frac{4}{3}$$

$$\Delta U = 900 = ?$$

ie process is isobaric

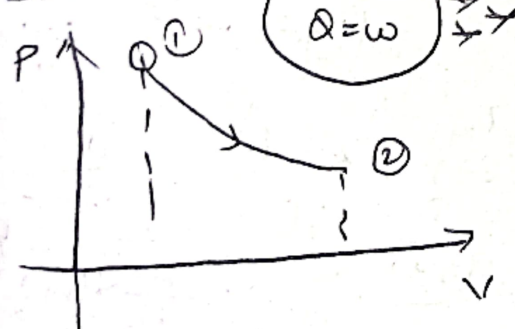
Isochoric Process

Volume constant

$$W = 0$$

$$Q = \Delta U = nC_v \Delta T$$

Isothermal Process



$$T = \text{constant}$$

$$P_V = \text{constant}$$

$$W = \int P dV = nRT \ln \left(\frac{V_2}{V_1} \right)$$

$$W = nRT \ln \left(\frac{P_1}{P_2} \right)$$

$$Q = 0$$

$$P V^\gamma = \text{constant}$$

$$T V^{\gamma-1} = \text{constant} \quad \text{--- (1)}$$

$$P^{1-\gamma} T^\gamma = \text{constant} \quad \text{--- (2)}$$

$$W = \frac{nR \Delta T}{1-\gamma}$$

$$W = -\Delta U$$

$P V^\gamma = \text{constant}$ Any Process

$$1. W = \frac{nR \Delta T}{1-\gamma} \quad \text{--- (1)}$$

$$2. \Delta U = nC_v \Delta T \quad \text{--- (1)}$$

$$3. Q = nC_p \Delta T \quad \text{--- (2)}$$

where

$$4. C = C_v + \frac{R}{1-\gamma}$$

C = Molar heat constant

Q

A diatomic Gas is undergoing process in which

$$T P^{5/2} = \text{constant} \quad T^2 = \frac{1}{P^5}$$

$$P T = \text{constant}$$

If Temp of Gas increased from T_0 to $3T_0$ find

$$1. W$$

$$2. \Delta U$$

$$3. Q$$

$$4. C = \frac{5R}{2} + \frac{R}{(1-\frac{2}{5})} = \frac{7}{5} + \frac{5}{2} = \frac{14}{5} + \frac{5}{2} = \frac{28}{10} + \frac{25}{10} = \frac{53}{10}$$

$$R V = n R T$$

$$P V = n R \left(\frac{1}{P^5} \right)$$